

Surrogate model motivation: MLP vs. GNN for per-member safety prediction

Flat MLP

~~connectivity discarded — nodes become anonymous numbers~~ ✕

$[x_1, y_1, x_2, y_2, \dots, x_n, y_n]$

flat coordinate vector — 78 values

Dense $\times 2$

ReLU · 256 units

Dense $\times 2$

ReLU · 128 units

Output — 1 value

global feasibility score

No structural correspondence:

force propagation follows the graph —

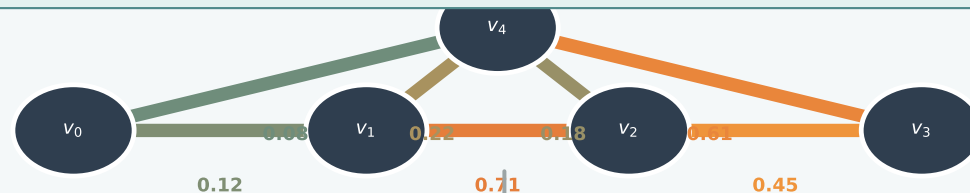
a flat vector carries no graph.

Indeterminate trusses require neighbour

context the MLP cannot access.

TrussEdgeSafetyGNN

graph topology preserved — each edge knows its neighbours



Node features: $[x, y, \text{support?}]$ · Edge features: $[L, A, E-I, \text{grade}]$

one feature vector per node; one per member

Message passing $\times 3$ layers

nodes aggregate neighbours $\rightarrow$ mirrors force propagation

Edge readout MLP

per-member hidden state $\rightarrow$ failure probability



120 members · per-member failure probability $\hat{p}_e$

